Theme: Pre-Delinquency Intervention Engine

VECTOR

Velocity engine for critical transaction and observation risk

-Detect Financial Stress Before It Becomes Default

Barclays Hack O Hire 2026

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Story: It Started With Sarah

Imagine Sarah, a model customer for six years, with a flawless repayment record and a strong relationship with her bank. Nothing seemed unusual until subtle changes began to appear. Her salary was credited five days late. Her savings dropped by 40%. Towards the end of the month, there were three unexpected ATM withdrawals.

Individually, these were small events. Together, they were early signals of financial stress.

Three weeks later, Sarah missed her payment. The system finally reacted. A red flag was raised and the collections team called. In a single interaction, a six-year healthy customer relationship was reduced to a recovery case. She felt judged, embarrassed, and disconnected from the bank she had trusted.

The tragedy is not that the risk appeared suddenly it didn't. Every warning sign already existed in the data. What was missing was a system that could interpret those signals early and act at the right moment.

During the pandemic, this was not just one Sarah. It was hundreds of thousands of customers, leading to £4.8 billion in bad-loan provisions—not because banks lacked data, but because they lacked anticipatory intelligence.

VECTOR changes that.

Introduction

The Pandemic Impact (2020 Barclays Data)

- **£3.7 billion** in credit impairment charges.
- Total bad loan provisions reached **~£4.8 billion.**
- Contributing to a **30-66% fall in profits.**

The Core Problem

- Data and technology existed, but systems weren't designed for early, personalized detection.
- Current systems are reactive (30+ days past due), leading to high costs and damaged relationships.

The Silent Signals of Stress

- Stress begins weeks before default with subtle, individual signals.
- Early warnings: Late salary, draining savings, late bills, unusual small withdrawals.
- Together, these form an unmistakable picture of impending default.

A New Fundamental Question

- The Pre-Delinquency Intervention Engine changes the focus:
- **NOT:** "How do we recover from default?"
- **BUT:** "How do we ensure it never happens?"

The Real Problem

Banks only look back. Never forward.

Most delinquencies can be avoided through a good proactive customer support by maintaining a strong customer relationship rather than a harsh collection team treatment.

By the time a payment is missed, it's already too late. Recovery rates drop to 40–60%. Collections costs shoot up by 15–20% of the recovered amount. And the customer who could have been helped with one early conversation is now a collections case.

Generic outreach makes it worse. The same message goes to everyone, ignored by over 80% of recipients. And when collectors call the damage to the customer relationship is often permanent.

The data to prevent all of this exists weeks before default. Current systems are simply not designed to read it.

Our Solution: Physics driven Anticipatory Intervention Engine

Traditional credit systems are inherently reactive, triggering only after a customer has already failed. Our **Anticipatory Intervention Engine** introduces a proactive paradigm shift by applying the **"Physics of Risk"** to banking data. By calculating the **Velocity** and **Acceleration** of behavioral signals, we detect a "Financial Death Spiral" weeks before a default occurs. This enables Barclays to transition from relationship damaging collections to human centered support saving millions in losses and preserving customer relationships.

Detailed Solution Breakdown

- **Multi-Expert Intelligence Architecture:** To handle fragmented data, our system utilizes a **Multi-Expert Ensemble**. Two specialized models, the **Financial Historian**, the **Behavioral Specialist** collaborate to provide a 360° view of customer health. This modularity ensures we pinpoint the exact "force" driving a customer's risk using static and dynamic features.

Our Solution: Physics driven Anticipatory Intervention Engine

- **Dynamic Trajectory Tracking (The Physics Engine):** We move beyond static, "look-back" scores to analyze dynamic customer trajectories. By measuring **Risk Velocity** (the speed of behavioral change) and **Risk Acceleration** (the rate at which a situation is worsening), we map every customer onto a **9-Zone Risk Matrix**. This identifies high-risk spirals even when current balances appear stable.
- **Stress Type Classifier:** After detecting a spiraling risk, providing a generic solution for every type of financial stress isn’t effective.Classifying it into stress categories so that GenAI can provide a meaningful intervene is a mandatory call. For this we use a dataset with labels augmented with domain knowledge.
- **GenAI-Powered Empathy (Intervene):** When the engine identifies the stress type, it replaces generic templates with **Generative AI** outreach. The system drafts hyper-personalized, empathetic messages matched to the customer's specific Risk Zone. Instead of a "Pay Now" alert, it proactively offers supportive solutions like "Payment Holidays" or restructuring plans in a tone that maintains long-term trust.

THE CORE LOGIC — THE PHYSICS OF RISK

Moving from Static Snapshots to Dynamic Trajectories

THE PROBLEM WITH TRADITIONAL SCORING: Traditional credit systems provide a static "look-back" score. They tell you where a customer was, but not where they are *going*. By the time a score drops, the customers would have entered a crisis.

OUR INNOVATION: VECTOR-BASED DETECTION We apply physics-based principles to 7 behavioral signals to detect a **"Financial Death Spiral"** weeks before a default occurs. We don't just measure debt, we measure the **momentum** of distress.

- **Risk Velocity (V):** Measures the **Speed of Change**. We track how fast a customer's signals are shifting (e.g., Credit utilization jumping 20% in 15 days). A high-velocity change indicates an immediate shift in financial stability.
- **Risk Acceleration (A):** Tracks the **Intensity of the Spiral**. This is the rate at which Velocity is increasing. Positive acceleration identifies a "Crisis Spiral"—the moment when spending frequency increases as savings decay exponentially.

Signal Framework

Instead of generic population rules, our engine monitors **7 high-frequency behavioral vectors** that represent the actual rhythm of a customer's financial life. Each signal is measured as a deviation from the user’s personal history to detect the earliest signs of distress.

- **Signal 1: Salary Timing (Drift)** – Measures the exact day salary arrives against the customer's 6-month median, delays are the earliest "Look-Ahead" indicators.
- **Signal 2: Income Erosion** – Tracks any month-on-month decrease in primary salary amount, identifying potential employment instability before it impacts bills.
- **Signal 3: Liquidity Momentum** – Monitors the speed of savings depletion; a week-on-week drop of >15% signals an exhausting "Safety Net."

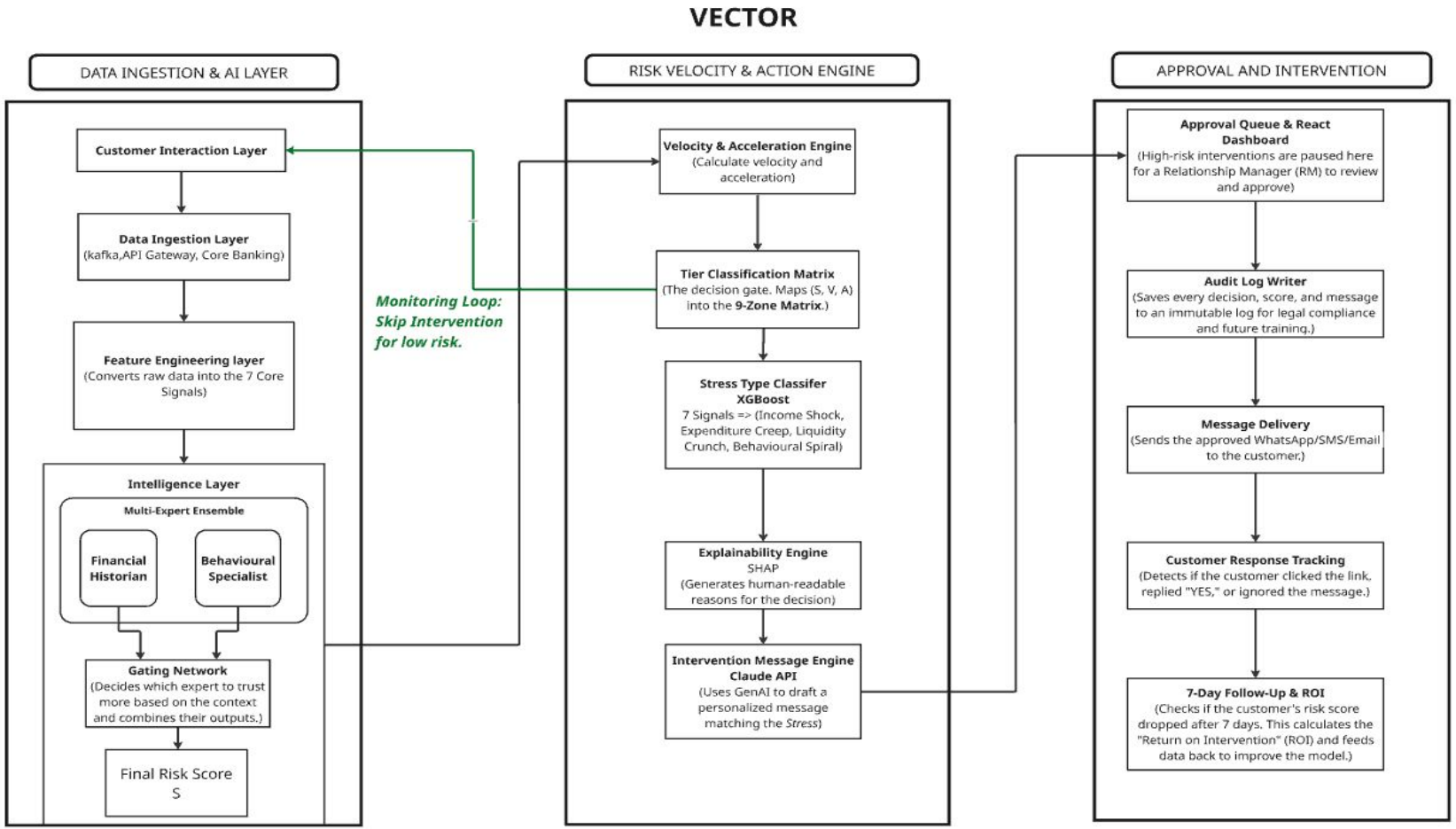
Signal Framework

- **Signal 4: Payment Integrity (Bill Drift)** – Detects when fixed obligations (utilities, rent) start shifting later in the monthly cycle, even if still paid.
- **Signal 5: Failed Auto-Debits** – Captures technical or balance-related bounces for EMIs and recurring subscriptions, marking the start of "Financial Juggling."
- **Signal 6: Credit Exhaustion** – Tracks the **Acceleration** of credit utilization; rapid jumps toward the limit indicate an emergency reliance on debt.
- **Signal 7: Engagement Decay (Digital Silence)** – Monitors psychological indicators such as avoiding the banking app or a sudden drop in budgeting tool usage.

Intervention Framework: Decision Matrix

Acceleration \ Velocity	Velocity Negative (Recovering)	Velocity Stable (Holding)	Velocity Positive (Deteriorating)
Accelerating (Worsening faster)	Zone 1 Automated 48hr Re-scan	Zone 2 Educational "Nudge" SMS	Zone 3 Immediate RM Intervention
Constant (Steady rate)	Zone 4 Log Outcome (Success)	Zone 5 Standard Monitoring	Zone 6 Scheduled Approval Queue
Decelerating (Slowing down)	Zone 7 Log & Validate Model	Zone 8 Monitor for Relapse	Zone 9 Second Intervention Attempt

Architecture Diagram



Tech Stack

CORE APPLICATION — MERN STACK

- **React.js (Frontend):** Relationship Manager dashboard with real-time transactions, risk metrics, and collection insights.
- **Node.js + Express (Backend):** Data orchestration, APIs, WebSocket live alerts, Intervention Approval Queue.
- **MongoDB (Database):** Stores 7 Behavioral Signals, risk snapshots, and flexible structured data.

INTELLIGENCE CORE — PYTHON MICROSERVICE

- **FastAPI:** High-speed async inference gateway for risk predictions.
- **Historian Model (XGBoost):** Structural financial risk analysis using historical credit data.
- **Behavioral Expert (LightGBM):** Detects behavioural risk signals from digital patterns.
- **Explainable AI (SHAP):** Converts predictions into clear decision explanations.

STREAMING & GENERATIVE LAYER

- **Kafka Streaming:** Real-time ingestion of transaction events (salary, ATM, bills).
- **GenAI (Claude/Gemini):** Generates personalized, empathetic outreach messages based on risk zone.

Data sources and characteristics

Dataset source:

1) **Bank Account Fraud Dataset Suite (NeurIPS 2022)**

2) **Home Credit Default Risk**

Data Reasoning:

The **Home Credit Default Risk** and **NeurIPS 2022 Bank Account Fraud (BAF)** datasets are considered as industry standards due to their robustness in analyzing **historical financial capacity** and **real time behavioral anxiety**, which are the two key factors in pre delinquency.

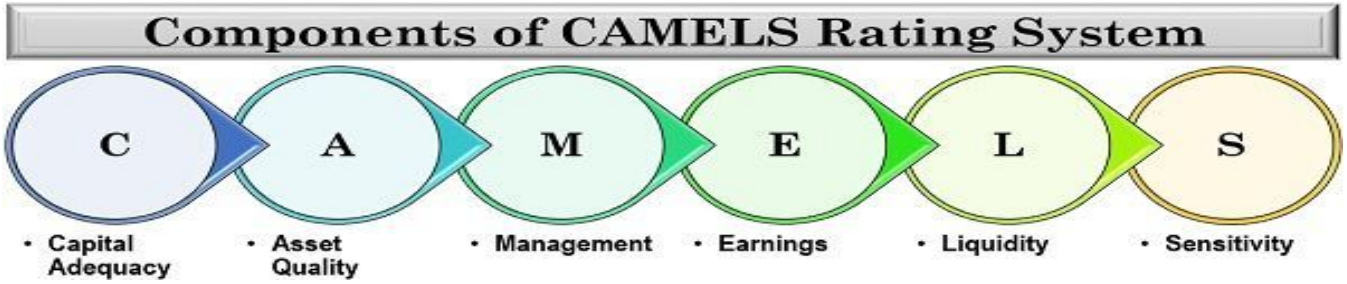
Framework used for validating data: CAMELS

The **CAMELS** framework is an internationally recognized supervisory rating system used by banking regulators (like the Federal Reserve and the RBI) to assess the overall condition, performance, and risk profile of financial institutions.

Data sources and characteristics

Why CAMELS ?

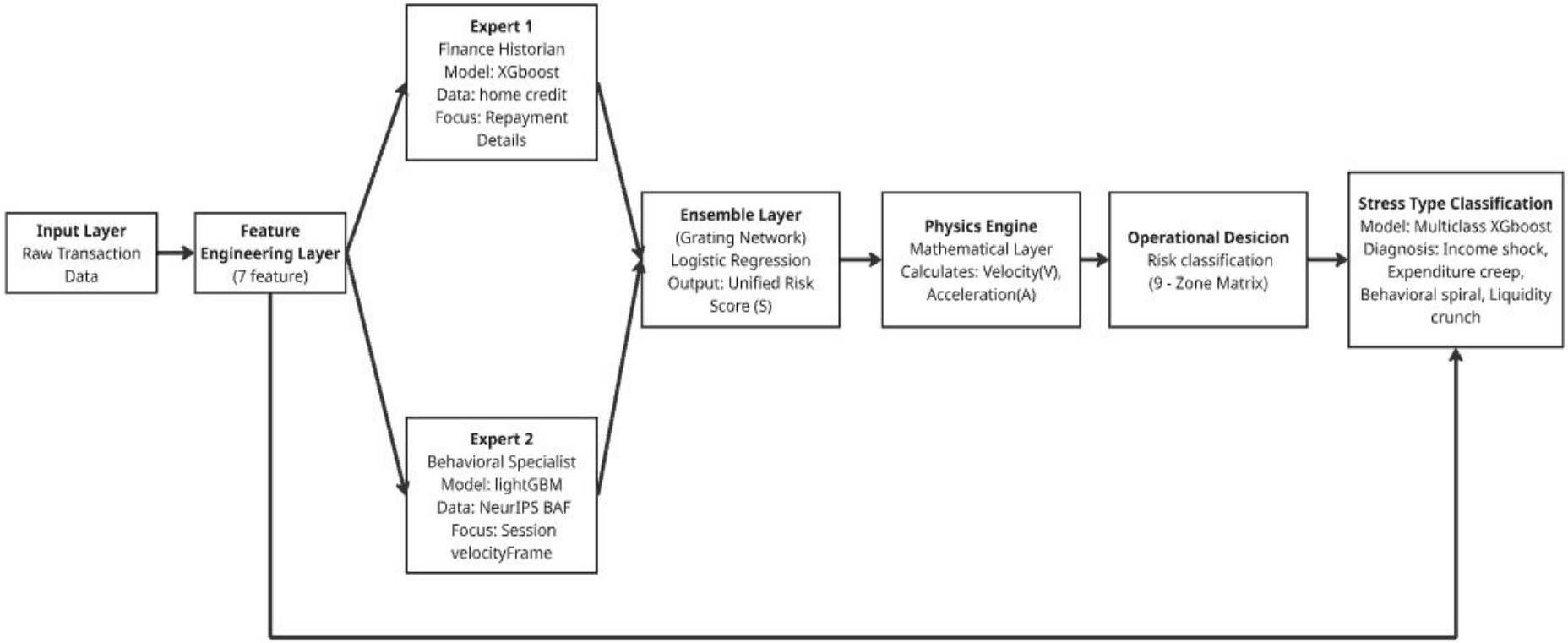
- **Regulatory Rigor :** It follows institutional risk-assessment standards
- **Strategic Precision:** Instead of sending a generic "Late Payment" notice, a tailored GenAI message can be sent based on the specific pillar that is failing (e.g., a "Liquidity" alert vs. a "Management" warning).
- **Holistic Risk Discovery**: Traditional models often rely on "lagging indicators" (static data). CAMELS requires a "qualitative and quantitative" approach that includes **Management** and **Sensitivity**. This framework justifies why you *must* add a behavioral dataset (BAF) to a static one (Home Credit).



Data source validation logic

CAMELS Pillar	Dataset Validation Logic	Supporting Features in Choice
C - Capital Adequacy	Measures the buffer to absorb losses. Home Credit is best because it identifies "unbanked" capital potential.	**Home Credit**: `AMT_INCOME_TOTAL`, `DebtRatio`.
A - Asset Quality	Evaluates the risk of the loan portfolio. Home Credit provides deep historical "asset decay" records.	**Home Credit**: `AMT_CREDIT_MAX_OVERDUE`, `num_tl_30dpd`.
M - Management	Assesses operational competency. BAF provides real-time "digital management" data that static sets lack.	**BAF**: `velocity_6h`, `velocity_24h` (Real-time control signals).
E - Earnings	Determines income stability and quality. Both datasets provide cross-validation of earning sustainability.	**Home Credit**: `MonthlyIncome`; **BAF**: `income` status.
L - Liquidity	Ability to meet short-term obligations. Home Credit's payment logs are essential for cash-flow integrity.	**Home Credit**: `installments_payments.csv` deltas.
S - Sensitivity	Resilience to market/psychological stress. BAF is the only "golden" source for digital session anxiety.	**BAF**: `session_length_in_minutes`, `keep_alive_session`.

Model Decision Architecture



Unique Selling Points

- **Physics-Based Risk Intelligence** – Uses Risk Velocity and Acceleration to track financial trajectory, not just static scores.
- **Early Detection Engine** – Identifies distress signals weeks before missed payments occur.
- **9-Zone Dynamic Risk Framework** – Maps customers into trajectory-based zones for precise, context-aware actions.
- **Multi-Expert AI Architecture** – Combines financial history, behavioral patterns, and digital engagement insights.
- **Explainable & Compliant Decisions** – Transparent AI reasoning with audit-ready explanations.
- **Human-in-the-Loop Governance** – Tiered intervention model ensuring safe automation with human oversight.
- **Empathetic Personalized Outreach** – Replaces generic collections messaging with tailored support actions.
- **Real-Time Intervention Pipeline** – Processes live data streams for immediate detection and response.
- **Prevention-Focused Strategy** – Shifts from reactive collections to proactive customer support.
- **Outcome-Driven Success Metrics** – Measures Zone Migration and customer recovery instead of just recovery rates.

Data Privacy & Minimization

We only store what is truly necessary, the eight key behavioral signals and essential metadata. We avoid storing full raw transaction descriptions unless absolutely required. This reduces risk and keeps the system focused on meaningful indicators rather than excess data.

- **Purpose Limitation**
  The data collected is used strictly for pre-delinquency intervention. It is not repurposed for marketing campaigns or unrelated profiling. The system operates within a clearly defined and ethical boundary.
- **Anonymised Data**
  Inside our machine learning pipelines, customer identities are protected. We use hashed customer IDs instead of direct personal information. There are no names, phone numbers, or identifiable details within the ML layer.
- **Encryption at Rest and In Transit**
  All stored data is secured using AES-256 encryption at rest. For communication between services and APIs, we use TLS 1.3 encryption to ensure secure data transfer across the network.
- **Data Retention Policy**
  Risk assessment snapshots are retained only for a defined time period — for example, 24 months. After that, the data is either automatically archived or permanently deleted according to policy.

Regulatory Compliance (FCA-Aligned Governance)

- **FCA Consumer Duty**
  VECTOR is built with strong awareness of FCA expectations, focusing on fair and responsible treatment of customers. The system is designed to identify early signs of financial stress and provide proactive support before harm occurs. Communication is kept clear, respectful, and empathetic, especially for vulnerable customers.
- **GDPR Compliance**
  The platform aligns with GDPR principles by ensuring transparency and user control over personal data. Customers have the right to understand how decisions are made through explainable AI methods such as SHAP. They can request access to their data, obtain explanations for risk assessments, and exercise the right to erasure where applicable. A Data Protection Impact Assessment (DPIA) is conducted to evaluate and manage privacy risks.
- **Model Governance**
  Strong governance practices ensure accountability and reliability. Model versions are tracked systematically, training datasets are documented, and bias and fairness testing is performed to reduce unintended discrimination. The system also follows a defined schedule for periodic recalibration to maintain performance and accuracy over time.
- **Human-in-the-Loop Safeguard**
  Critical decisions always involve human oversight. High-risk actions require approval from a relationship manager, and no autonomous restructuring occurs without review. A clear escalation pathway is defined to ensure responsible handling of sensitive situations.

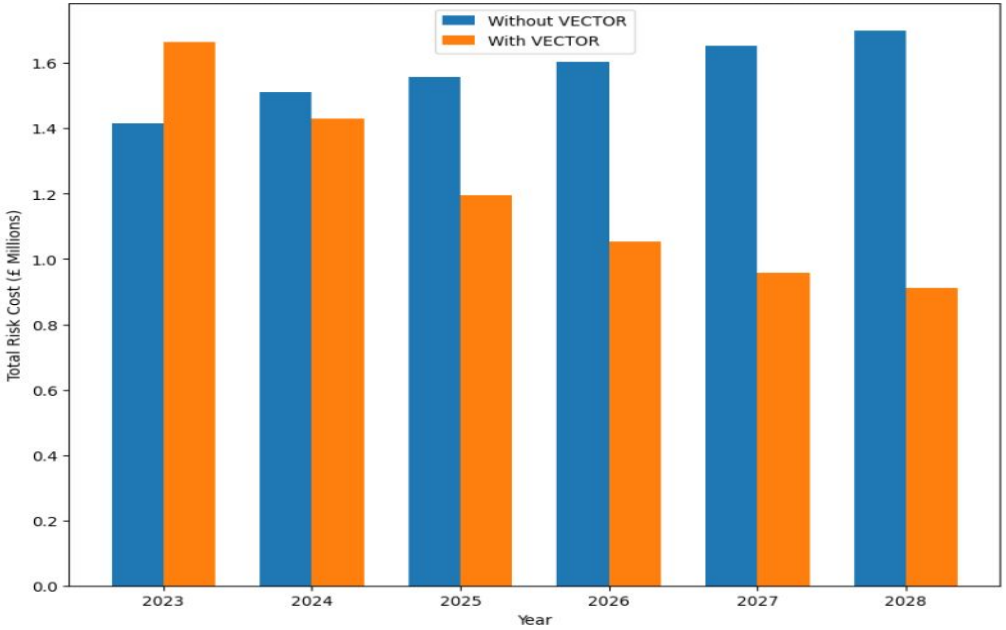
Business Impact

Key Observations

- Data from **2023–2025** represents original historical values.
- Data from **2026–2028** is predicted based on our model projections.
- Without VECTOR, total risk cost shows a steady upward trend, indicating increasing financial exposure.
- With VECTOR, projected data shows a consistent decline in risk cost due to proactive intervention and predictive risk management.

Strategic Impact

- Reduced credit losses..
- Improved prediction accuracy and early risk detection.
- Better resource allocation and operational efficiency.
- Strong long-term financial stability and growth.



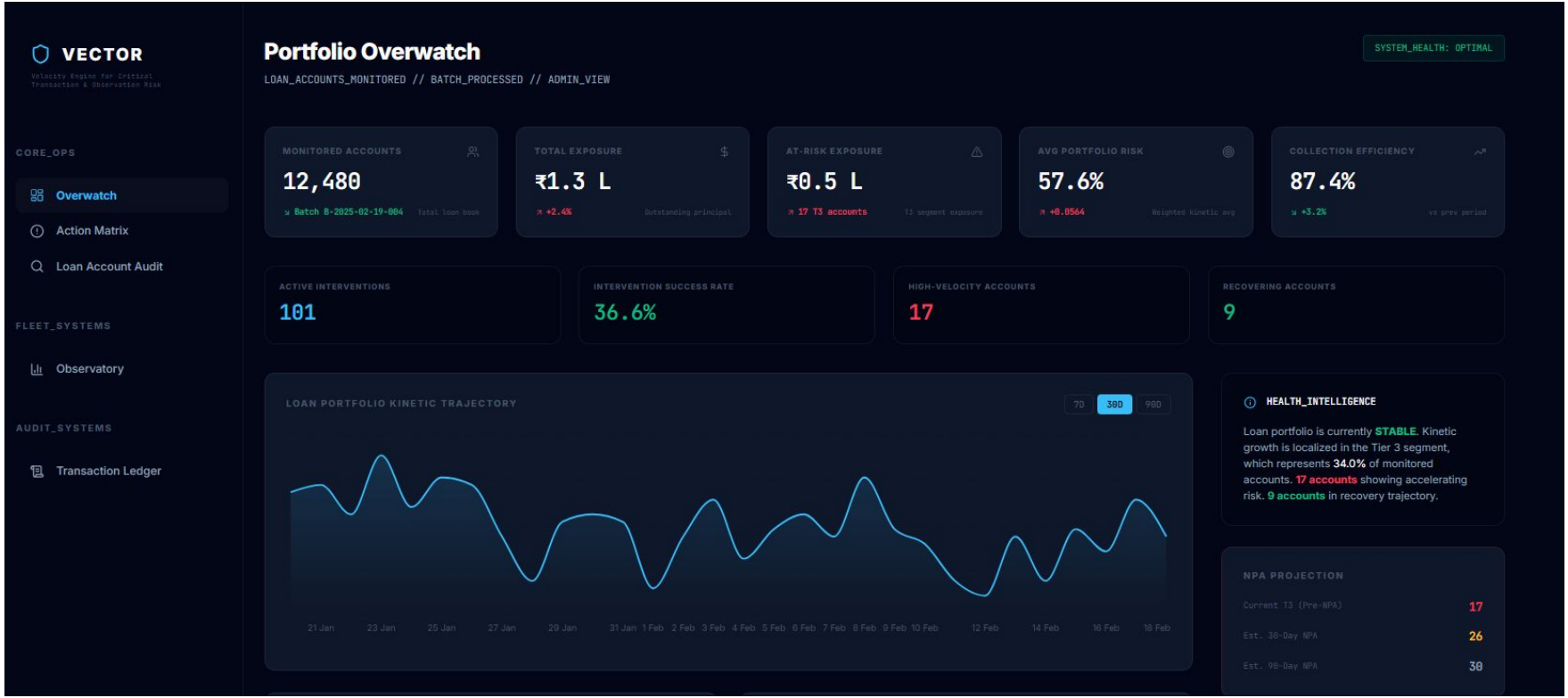
Overall Inference

VECTOR shifts risk management from reactive handling to predictive prevention, resulting in measurable cost savings and improved business outcomes.

Prototype

Github : Repo Link

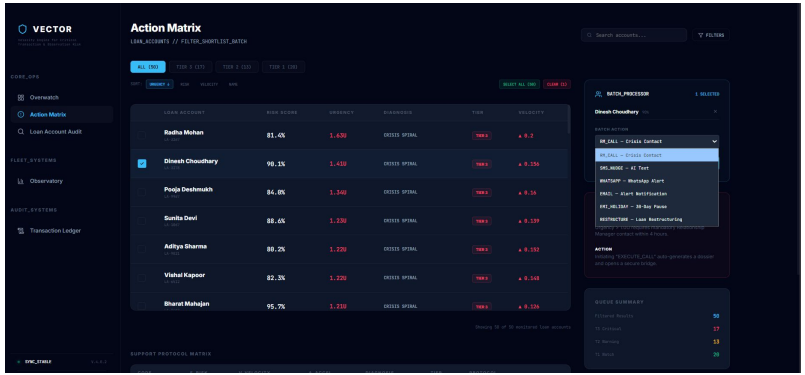
Demo - Homepage

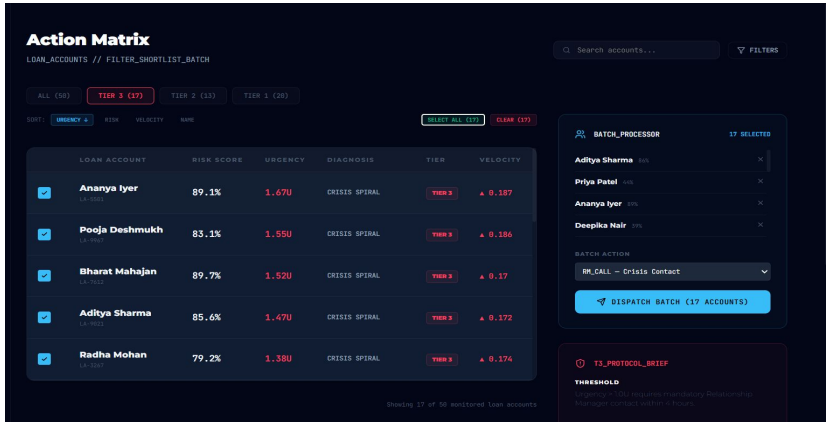


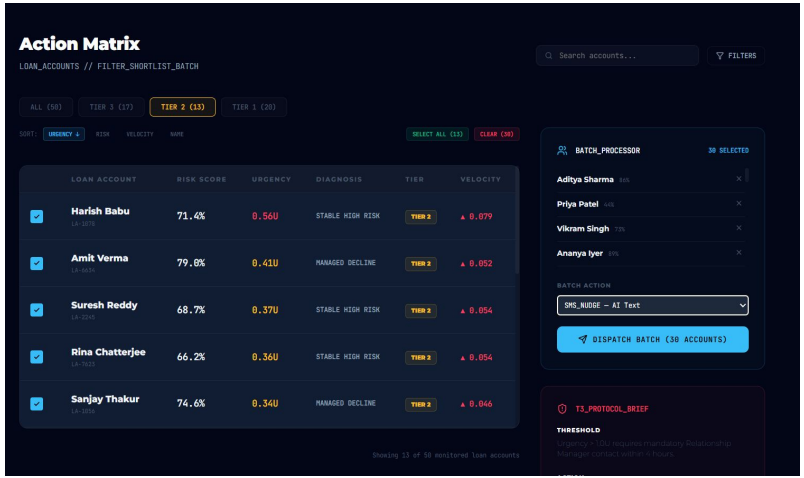
Demo - Homepage

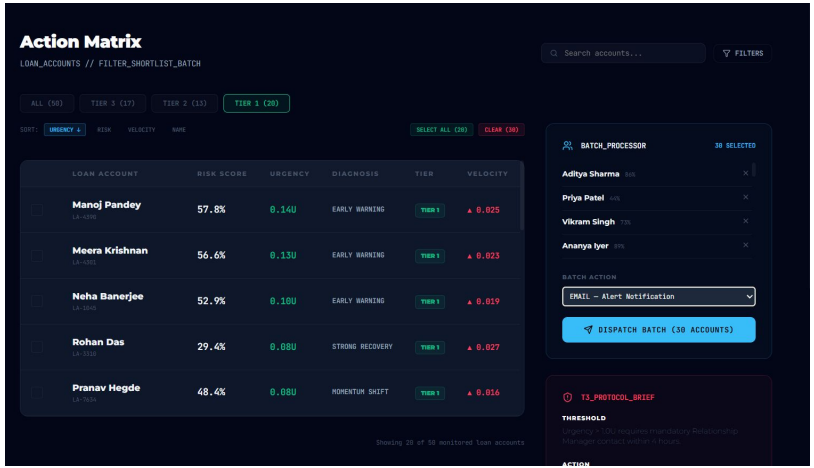
Demo - Customer List

Demo - Action Matrix

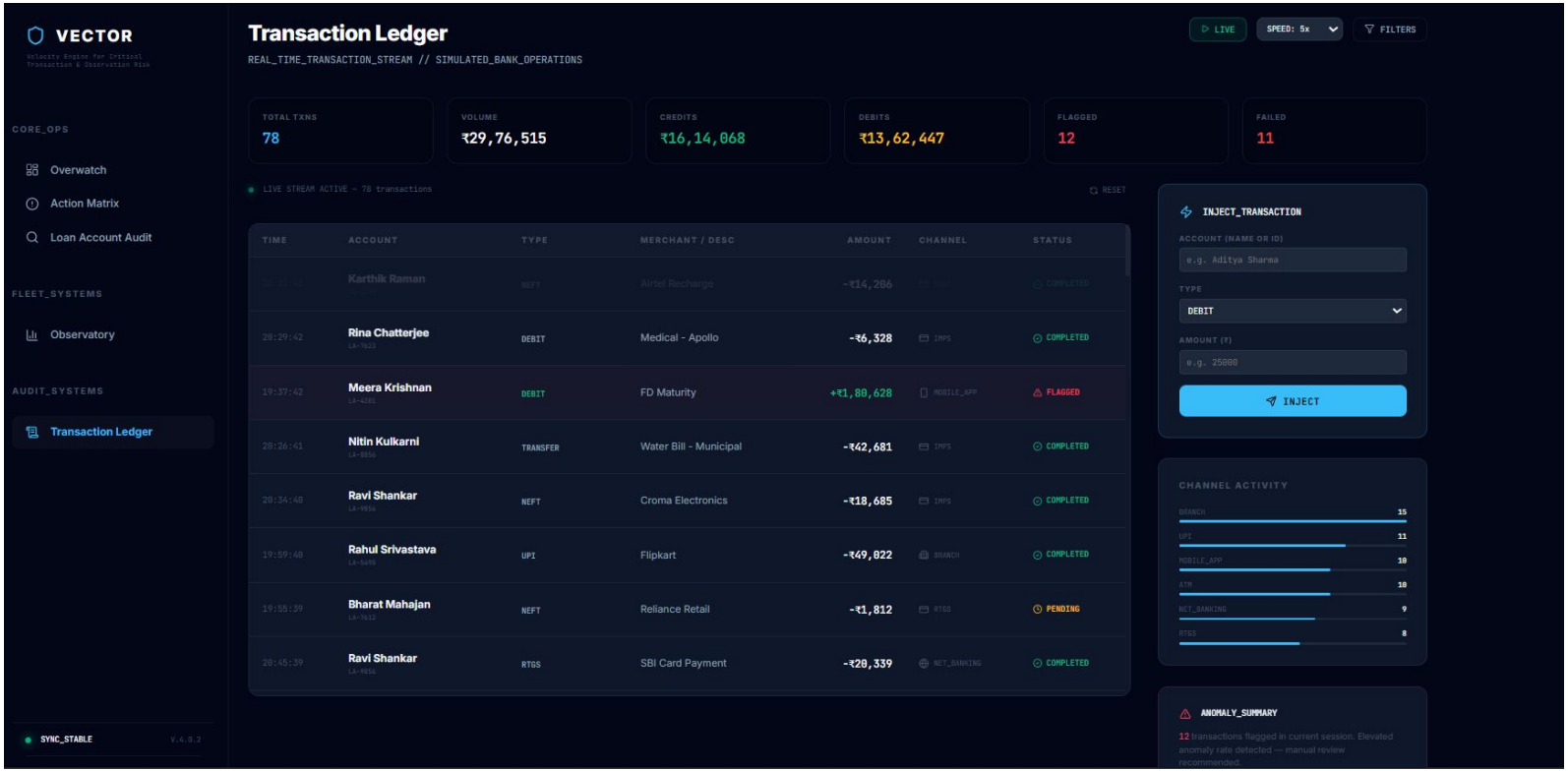








Demo - Overall Ledger



Future Scope

- **Agentic AI Decision Layer**
  Introduce autonomous AI agents that continuously monitor risk trajectory, evaluate multiple intervention strategies, and recommend optimal actions dynamically based on customer context and historical outcomes. Still preserving Human in the loop policy.
- **Adaptive Decision Orchestration**
  Move from rule-based triggers to goal-driven AI agents capable of prioritizing actions (payment rescheduling, proactive outreach, risk monitoring) while maintaining human-in-the-loop governance for high-impact decisions.
- **Post-Intervention Intelligence & Feedback Loop**
  Track customer response after intervention (behavior change, repayment stability, engagement signals) and feed outcomes back into the model to continuously improve intervention effectiveness.
- **Closed-Loop Risk Lifecycle Management**
  Extend VECTOR from early detection → decision → intervention → outcome tracking → continuous learning, creating a fully adaptive anticipatory banking ecosystem.

Conclusion

VECTOR represents a fundamental shift in banking intelligence. By moving from **static snapshots** to **dynamic trajectories**, we solve the industry's greatest challenge: the **Reactivity Gap.**

The VECTOR Edge:

- **Predictive Precision:** By calculating the **Velocity (V)** and **Acceleration (A)** of risk, we identify the "Death Spiral" of financial distress **weeks before** a traditional credit score moves.
- **Synchronized Intelligence:** Our dual-expert architecture bridges the gap between **historical stability** (Home Credit) and **real-time digital friction** (BAF), ensuring every prediction is grounded in both context and current behavior.
- **Ethical Innovation:** Through **Privacy-by-Design** and **Explainable AI**, we transform automated monitoring into a supportive, human-led partnership. We don't just flag failures; we provide the relationship managers with the "Why" and the "How" to fix them.

VECTOR evolves Barclays from a reactive lender into a **Proactive Financial Guardian**. We reduce operational losses and NPLs, but more importantly, we preserve **Lifetime Customer Value**.

"SUCCESS IS NO LONGER MEASURED BY HOW MUCH DEBT WE RECOVERED, BUT BY HOW MANY TRAJECTORIES WE CORRECTED"

Thank You